

The diagram illustrates the process of DNA replication at a replication fork. A vertical red dashed line represents the fork. To the left, the template strand is oriented 3' to 5' (top to bottom), and the new strand is synthesized continuously towards the fork (5' to 3' from top to bottom). To the right, the template strand is oriented 5' to 3' (top to bottom), and the new strand is synthesized discontinuously as Okazaki fragments away from the fork (5' to 3' from bottom to top). Key features include the RNA primer (green), DNA polymerase III (grey), and the RNA primer being removed by DNA polymerase I (grey).

Labels and Components:

- AC**: Nucleotide sequence at the top of the leading strand.
- AG**: Nucleotide sequence at the top of the lagging strand.
- ATTCTT**: RNA primer sequence on the lagging strand.
- A**: Nucleotide sequence on the lagging strand.
- C**: Nucleotide sequence on the lagging strand.
- A**: Nucleotide sequence on the lagging strand.
- A**: Nucleotide sequence on the lagging strand.
- TTCCTCTA**: RNA primer sequence on the lagging strand.
- T**: Nucleotide sequence on the lagging strand.
- GT**: Nucleotide sequence on the lagging strand.

The diagram illustrates a DNA microarray with 20 rows. Each row consists of two green horizontal bars. The left bar is labeled 'TTCCTCTA' and the right bar is also labeled 'TTCCTCTA'. A vertical dashed red line is positioned between the two bars of each row, running through the center of the entire array.

The diagram illustrates a DNA double helix structure. A vertical red dashed line is positioned between the two strands, indicating a site of mutation or repair. The sequence **TTCCTCTA** is highlighted in green on both strands. The strands are composed of gray rectangular blocks representing nucleotides. The top strand (left) has a 'G' at the second position from the left. The bottom strand (left) has a 'C' at the second position from the left. The top strand (right) has an 'A' at the second position from the left and an 'A' at the fourth position from the left. The bottom strand (right) has a 'T' at the second position from the left and a 'T' at the fourth position from the left. The sequence **TTCCTCTA** is highlighted in green on both strands.

Reporter position